

HEALTHCONNECT — WEEK 7

Cross-Track Coordination Record

Data Analytics ↔ Data Science

The DAX Critical_Risk_No_Show_Rate, Executive Attendance Dashboard and da_ds_feature_specs.csv were tested together. The initial DAX result was 60.00% rather than the expected 61.76% because cancelled appointments were included. Feature ingestion also failed because of missing reminder values. The team added an explicit cancellation exclusion and imputed reminder_channel as No Reminder. Retest produced 61.76% across slicer states and successful ingestion of all 5,000 records.

ML Engineering ↔ Analytics context

The integrated ML pipeline was tested for valid inputs, missing fields, invalid numeric values, unseen categories, feature engineering, preprocessing, batch prediction, model reload, configurable threshold and reproducibility. The supplied Week 7 testing summary reports 13/13 tests passed after refinements.

Data Science handoff validation

The Week 6 RiskFeature Data Science Handoff was checked for duplicate appointment IDs, risk-score/tier consistency and missing distance values. There were 0 duplicate IDs, 0 risk-score/tier mismatches and 86 missing distance values (1.8%). A distance_missing indicator was added and the corrected handoff was re-exported.

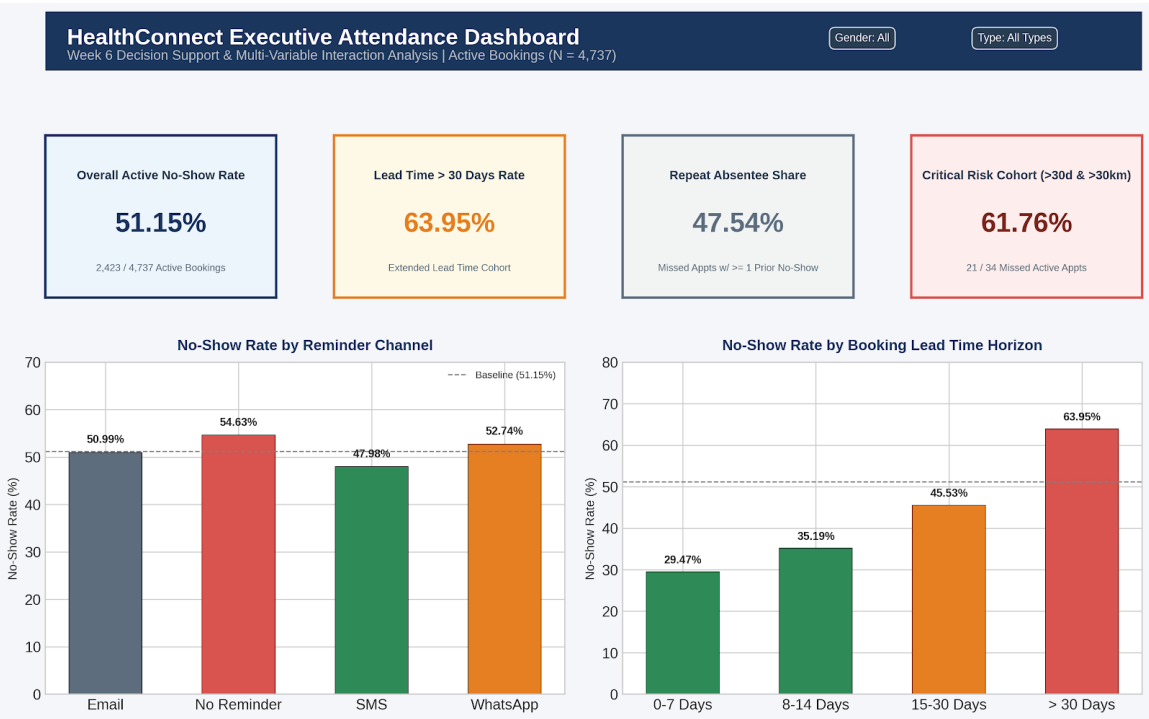
PM evidence control

The record uses the actual testing evidence supplied. Communication alone is not treated as a testing result. Where a dependency remains unresolved, it is recorded for Week 8.

Remaining cross-track dependencies

Final predicted probabilities for Power BI risk filtering; corrected handoff adoption confirmation; operational threshold validation; ML logging follow-up; reminder-timing data limitation; and review of the n=13 interaction.

Supporting visual evidence



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INFO | Received prediction request for 1 record(s).
ERROR | Input validation failed: Numeric range validation failed: 'previous_no_shows' has 1 value(s) outside [0, 10000]
INFO | Received prediction request for 1 record(s).
INFO | Prediction succeeded. mean_probability=0.4434, n_flagged=0

negative_age:
  PASS - rejected with: Numeric range validation failed: 'age' has 1 value(s) outside [0, 120]

negative_booking_lead_days:
  PASS - rejected with: Numeric range validation failed: 'booking_lead_days' has 1 value(s) outside [0, 365]

negative_distance_to_clinic_km:
  PASS - rejected with: Numeric range validation failed: 'distance_to_clinic_km' has 1 value(s) outside [0, 500]

negative_previous_no_shows:
  PASS - rejected with: Numeric range validation failed: 'previous_no_shows' has 1 value(s) outside [0, 10000]

Valid record regression check:
  Prediction: 0
  Probability: 0.4434
  Valid prediction: True

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PASS: All invalid-input scenarios were rejected correctly.
PASS: Valid-input prediction still works after refinement.
PASS: Week 7 input validation refinement passed retest.

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Figure 3 — ML Invalid-Input Validation and Regression Retest

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ML-02 - MISSING REQUIRED FIELD TEST
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Dropped column: 'distance_to_clinic_km'
Remaining columns: ['gender', 'age', 'age_group', 'appointment_type', 'appointment_day', 'appointment_time', 'booking_lead_days', 'previous_appointments', 'previous_no_shows', 'reminder_sent', 'reminder_channel']

Exception raised: True
Error detail: KeyError: 'distance_to_clinic_km'

PASS: ML-02 - pipeline correctly rejected an incomplete input.

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Figure 4 — ML Missing Required Field Test

HealthConnect Clinic — Executive Analytics & Testing Dashboard



Figure 5 — HealthConnect Clinic Executive Analytics & Testing Dashboard